By Maha Rana

IT601P Updated Current Paper Solved

LDAP (Lightweight Directory Access Protocol) Steps?

1. **Install LDAP Server**

- On **Ubuntu/Debian**:

```
bash

sudo apt update
sudo apt install slapd ldap-utils
```

On **Red Hat/CentOS**:

```
bash

sudo yum install openldap openldap-servers openldap-clients
```

2. **Configure LDAP Database:**

- Set the **organization name** and **domain** (e.g., `dc=example,dc=com`).
- Use `dpkg-reconfigure slapd` (on Debian-based systems) to reconfigure.
- Modify the configuration files (`/etc/ldap/ldap.conf` or `/etc/openldap/slapd.conf`).

3. **Start and Enable LDAP Service:**

```
bash

sudo systemctl enable slapd
sudo systemctl start slapd
```

4. **Verify LDAP Installation**

```bash
ldapsearch -x -LLL -b "dc=example,dc=com"
```

5. **Create LDAP Users and Groups**

   Define **Organizational Units (OU)**, **Users**, and **Groups** in an LDIF file:

```ldif
dn: ou=users,dc=example,dc=com
objectClass: organizationalUnit
ou: users
```

Load the configuration using:

```bash
ldapadd -x -D "cn=admin,dc=example,dc=com" -W -f users.ldif
```

6. **Set User Authentication (if needed)**

- Configure clients (Linux or Windows) to authenticate against the LDAP directory.
- Modify **PAM (Pluggable Authentication Module)** or `/etc/nsswitch.conf` for Linux authentication.

- **Integrate LDAP with Other Services**
  - LDAP can be used for authentication in **SAMBA, Apache, OpenVPN, Mail Servers (Postfix), etc.**

2. X –b –h ?

Answer:

The **`x -h -b`** command syntax does not directly relate to a standard Linux or LDAP command. However, if you're referring to **LDAP search commands (`ldapsearch`)**, here's what the correct parameters mean:

Correct Syntax for LDAP Search:

```bash
ldapsearch -x -h <hostname> -b "<base_DN>"
```

Explanation of Parameters

- `-x` → **Use simple authentication** (instead of SASL).
- `-h <hostname>` → **Specify the LDAP server hostname** (e.g., `ldap.example.com`).
- `-b "<base_DN>"` → **Base Distinguished Name (DN)** to search from (e.g., `"dc=example,dc=com"`).

Maha Rana

Enable LDAP Steps on Linux?

To enable and configure LDAP on a **Linux system**, follow these steps:

Step 1: Install LDAP Server

For **Ubuntu/Debian**:

```bash
sudo apt update
sudo apt install slapd ldap-utils
```

Step 2: Configure LDAP Server

- Run the LDAP configuration command:

```bash
sudo dpkg-reconfigure slapd  # Ubuntu
```

- Define **Base DN** (e.g., dc=vusna,dc=com).
- Set the **Admin password** and confirm the settings.

Step 3: Verify LDAP Service

Start and enable LDAP:

```bash
sudo systemctl start slapd
sudo systemctl enable slapd
```

Check if LDAP is running:

```bash
sudo systemctl status slapd
```

Step 4: Add Organizational Units (OU), Users, and Groups

Create an LDIF file (`vusna.ldif`) with:

```ldif
dn: ou=users,dc=vusna,dc=com
objectClass: organizationalUnit
ou: users
```

Load the configuration:

```bash
ldapadd -x -D "cn=admin,dc=vusna,dc=com" -W -f vusna.ldif
```

Step 5: Configure Client Machines (Optional)

- Install libnss-ldap and libpam-ldap for Linux authentication.
- Modify /etc/nsswitch.conf and /etc/ldap/ldap.conf for proper integration.

2. Steps to Configure SSHD (SSH Daemon) in Linux?

Step 1: Install OpenSSH Server (if not installed)

For **Ubuntu/Debian**:

```bash
sudo apt install openssh-server -y
```

For **CentOS/RHEL**:

```bash
sudo yum install openssh-server -y
```

Step 3: Modify SSH Configuration File

Edit the SSH configuration file:

```bash
sudo nano /etc/ssh/sshd_config
```

Change the SSH port (optional):

```yaml
Port 2222
```

Disable root login for security:

```nginx
PermitRootLogin no
```

Allow only specific users (e.g., `admin`):

```nginx
AllowUsers admin
```

Step 4: Restart SSH Service:

```bash
sudo systemctl restart ssh
```

Step 5: Test SSH Connectivity

From another system:

```bash
ssh user@server-ip
```

3. Use of |, <, >> in Linux?

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Symbol	Description	Example
`	` (Pipe)	Passes output of one command as input to another
`<` (Input Redirection)	Reads input from a file instead of keyboard	`cat < file.txt`
`>>` (Append Output)	Appends output to an existing file	`echo "Hello" >> output.txt`

4. Output Prediction (Example Questions)?

Question 1:

```bash
echo "Hello World" | wc -c
```

Output:

```bash
12  # (Includes newline character)
```

Question 2:

```bash
cat < file.txt
```

Output:

- The word `"Linux"` is appended to `myfile.txt` and displayed.

5. Solve `ldap@vusna.com` with IP Address?

- Assume the **LDAP server is running at** `192.168.1.10` for the domain `vusna.com`.

LDAP Query Command Example:

```bash
ldapsearch -x -h 192.168.1.10 -b "dc=vusna,dc=com"
```

This will fetch **all LDAP entries** from the server?

Adding a User to LDAP (`ldapadd`)

Create a user file (`user.ldif`):

```ldif
dn: uid=user1,ou=users,dc=vusna,dc=com
objectClass: inetOrgPerson
uid: user1
cn: User One
sn: One
userPassword: password123
```

Add the user with:

```bash
ldapadd -x -D "cn=admin,dc=vusna,dc=com" -W -f user.ldif
```

Modify LDAP User (`ldapmodify`)

To change the user's password:

```bash
ldappasswd -x -D "cn=admin,dc=vusna,dc=com" -W -S "uid=user1,ou=users,dc=vusna,dc=com"
```

Summary

- **LDAP Steps**: Install, configure, add users/groups, verify.
- **SSHD Steps**: Install, configure `/etc/ssh/sshd_config`, restart, test.
- **Linux Operators (`|`, `<`, `>>`)**: Used for piping, redirection, and appending.
- **LDAP Commands (`ldapsearch`, `ldapadd`, `ldapmodify`)** for managing `ldap@vusna.com`.

1. Explanation of |, >, >> Symbols?

Symbol	Description	Example
`	` (Pipe)	Passes output of one command as input to another.
`>` (Output Redirection)	Overwrites output to a file.	`echo "Hello" > file.txt`
`>>` (Append Output)	Adds output to an existing file.	`echo "New Line" >> file.txt`

2. Explanation of -x, -LLL, -H, -b in LDAP Commands?

Option	Description	Example
`-x`	Uses simple authentication instead of SASL.	`ldapsearch -x`
`-LLL`	Removes unnecessary info from LDAP output.	`ldapsearch -x -LLL`
`-H`	Specifies the LDAP server URL.	`ldapsearch -x -H ldap://192.168.1.10`
`-b`	Sets the search base DN (Distinguished Name).	`ldapsearch -x -b "dc=example,dc=com"`

3. Writing Commands for Given Email Output?

(Assuming the email-related output required extracting email addresses from LDAP)

Command to list all email addresses in LDAP:

```bash
ldapsearch -x -LLL -b "dc=example,dc=com" mail
```

Command to filter a specific user's email:

```bash
ldapsearch -x -LLL -b "dc=example,dc=com" "(uid=johndoe)" mail
```

4. Configuring Host Based on Given Location and Address

(Assuming the location and address were related to `/etc/hosts` file configuration)

Example: If the given address is **`192.168.1.20`** and hostname is **`server1.example.com`**, add this entry

```bash
echo "192.168.1.20 server1.example.com server1" | sudo tee -a /etc/hosts
```

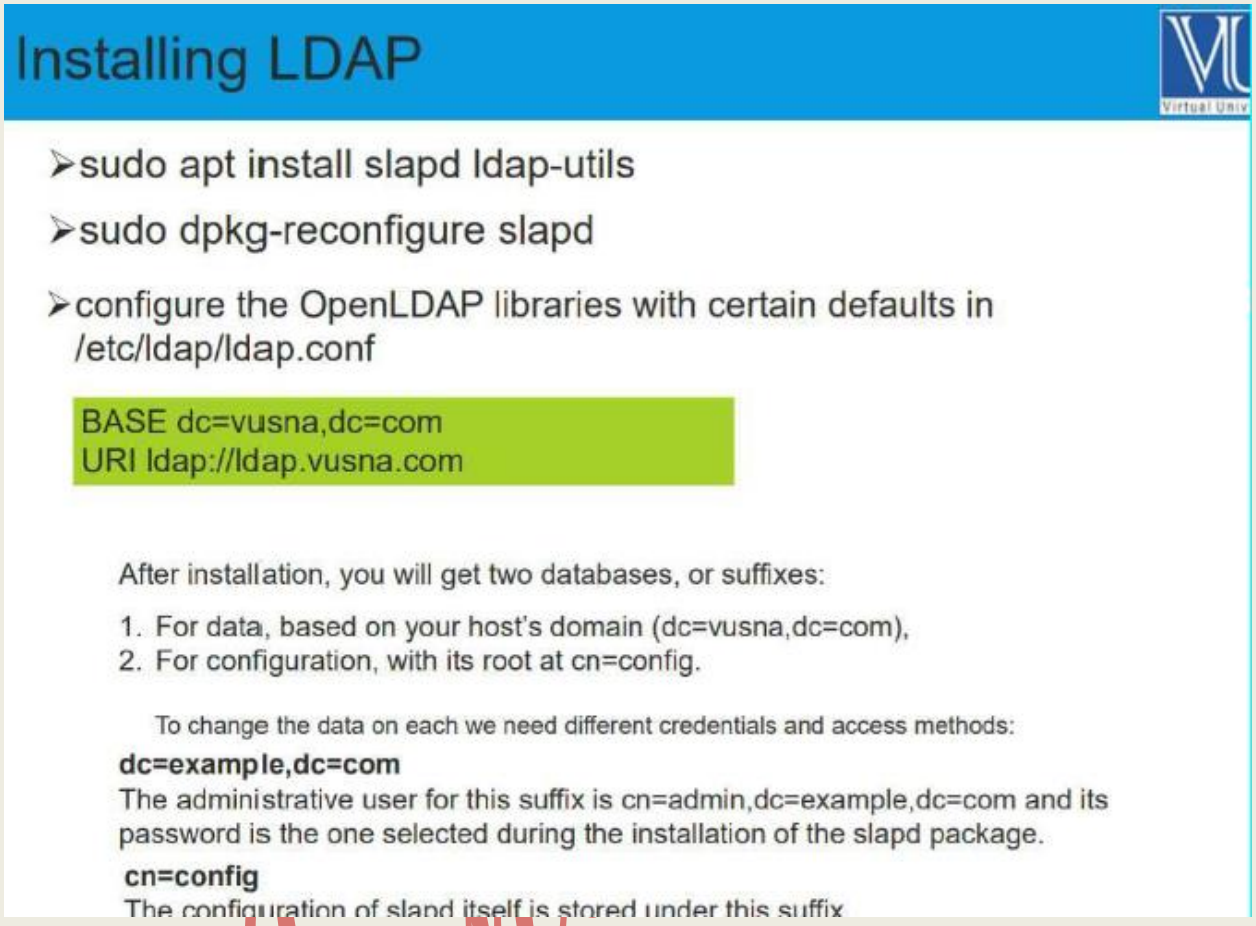
To verify:

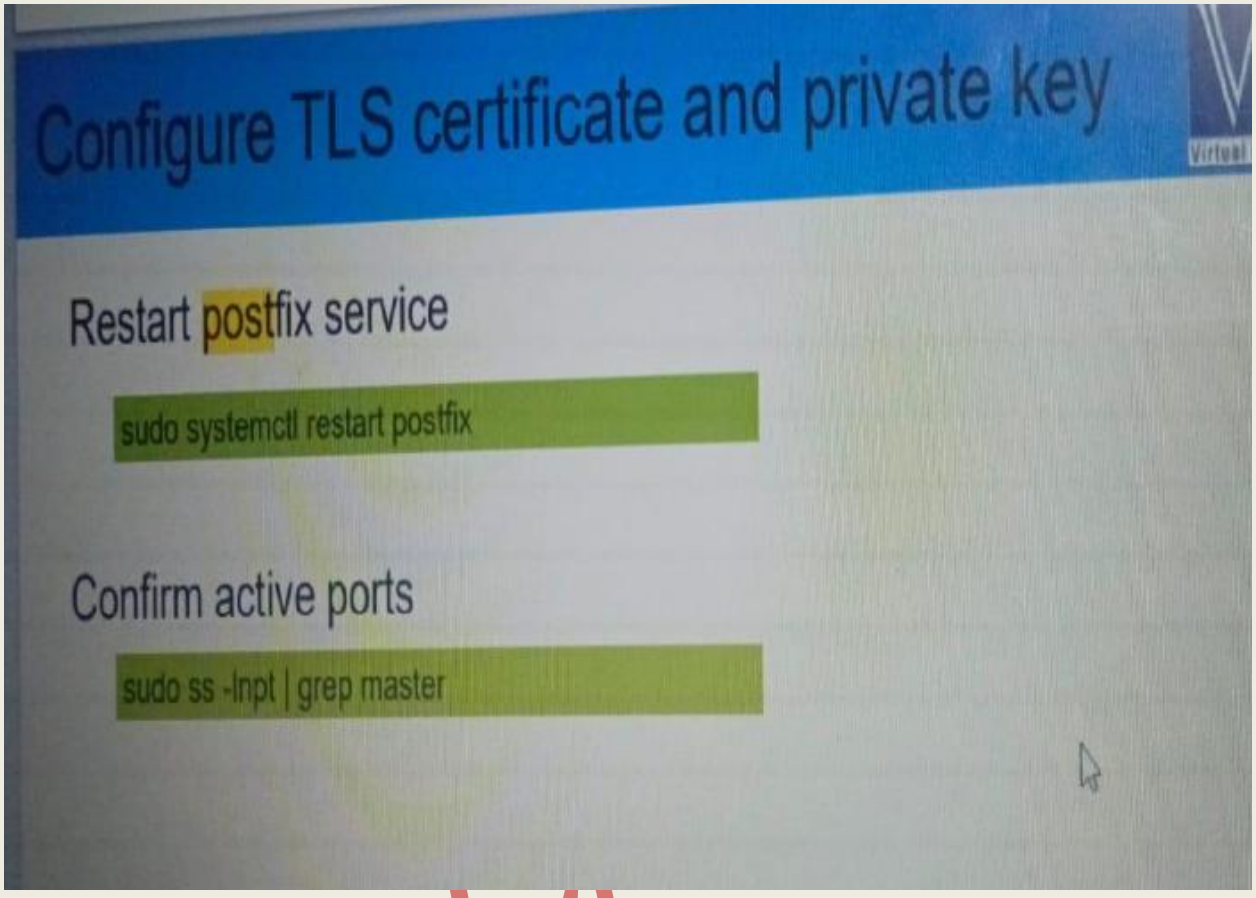
```bash
cat /etc/hosts
```

Summary:

- `|`, `>`, `>>` used for piping and output redirection.
- LDAP flags `-x`, `-LLL`, `-H`, `-b` explained with examples.
- Email filtering command provided.

- Host configuration command added.




Configure TLS certificate and private key
Restart postfix service
sudo systemctl restart postfix
Confirm active ports
sudo ss -lnpt | grep master

Maha

we required three types of dependencies

1) cs-certificates

2) curl

3) gnupg

There are two types keys

1) public key

2) private key

Inko ham authentication or encryption k Liya use kar sakty

Ca certificates are electronic passwords jinko ham authentication and or encryption k Liya use karty hy

ca certificates k Liya Jo certified authority hoti hy os k public certificates ko manage karny k Liya ca-certificate ka package use hota hy isko different server communication ya identity verification k Kiya b use kar sakty hy

4. install ca-certificate
   - it601p@labserver: ~$ sudo apt install ca-certificate

mahjabeen97869@gm

curl package is a client url Jo k diff web server k sath communication k Liya use hota hy eg file download ya server sy response retrieve karny k liya

5. install curl package
   - it601p@labserver: ~$ sudo apt install curl

next package is pretty good privacy gnu pgp isko ham different files ko encrypt karny k Liya use Kary gy ya fr hmari communication ko sign ya is sy ham apni or remote server ki identity ko verify kar saky gy isko ham ssh k sath b use kar sakty

6. install gpg
   - it601p@labserver: ~$ sudo apt install gnupg

These are some of our initial requirements before installing docker

Now we have to install the keys of docker in our system

0755 mean world/ globally readable

7. To change the permission of directory
   - it601p@labserver: ~$ sudo install-m 0755 -ld/etc/apt/keyrings

mahareen97869@gmail.com

Installation and Configuration

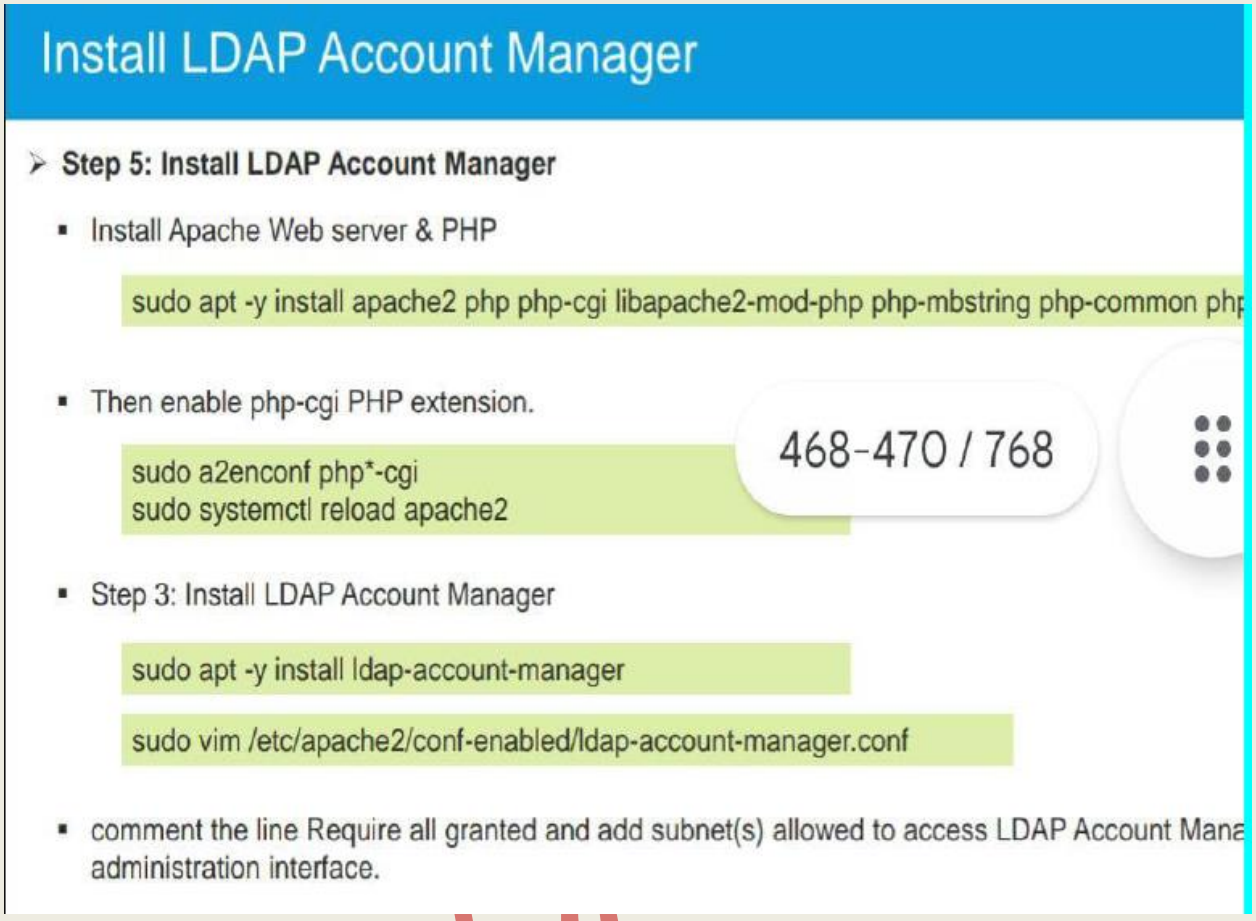
- **Step 1: Set hostname for the Ubuntu server**
- **Step 2: Install OpenLDAP Server**

```
sudo apt update
sudo apt -y install slapd ldap-utils
```

Verify installation with command slapcat to output SLAPD database contents.

- **Step 3: Add base dn for Users and Groups**
  - Add a base DN for users and groups. Create a file named basedn.ldif and the file by running tl ldapadd
- **Step 4: Add User Accounts and Groups**
  - Generate a password for the user account to add.
  - Create ldif file for adding users.
  - When done with edit, add account by running ldapadd command.
  - Create ldif file for groups and add groups with command ldapadd

Maha


Install LDAP Account Manager
Step 5: Install LDAP Account Manager
Install Apache Web server & PHP
sudo apt -y install apache2 php php-cgi libapache2-mod-php php-mbstring php-common php
Then enable php-cgi PHP extension.
sudo a2enconf php*-cgi
sudo systemctl reload apache2
468-470 / 768
Step 3: Install LDAP Account Manager
sudo apt -y install ldap-account-manager
sudo vim /etc/apache2/conf-enabled/ldap-account-manager.conf
comment the line Require all granted and add subnet(s) allowed to access LDAP Account Mana
administration interface.

Maha

Installing NFS on Linux Server

- **Install the nfs-kernel-server software package.**
  - sudo apt install nfs-kernel-server
- **Start the NFS server**
  - sudo systemctl start nfs-kernel-server.service
- **Configure the directories to be exported by adding them to the /etc/exports file.**
  - /srv *(ro,sync,subtree_check)
  - /cshare *.vusna.com(rw,sync,no_subtree_check)
  - /scratch *(rw,async,no_subtree_check,no_root_squash)
- **Apply the new config via**
  - sudo exportfs -a
- **Check status or restart service**
  - sudo systemctl restart nfs-kernel-server

- **rw**
  - It gives the client computer both read and write access to the volume.
- **sync**
  - It forces NFS to write changes to disk before replying. This results in a more stable and consistent environment but reduces the speed of file operations.
- **no_subtree_check**
  - It prevents subtree checking, which is a process where the host must check whether the file is actually still available in the exported tree for every request. This can cause many problems when a file is renamed while the client has it opened. In almost all cases, it is better to disable subtree checking.
- **no_root_squash**
  - By default, NFS translates requests from a root user remotely into a non-privileged user on the server. This was intended as security feature to prevent a root account on the client from using the file system of the host as root. no_root_squash disables this behavior for certain shares.

Configure TLS certificate and private key

Restart postfix service

```
sudo systemctl restart postfix
```

Confirm active ports

```
sudo ss -lnpt | grep master
```

Install IMAP and POP3 Server

- Installation

```
sudo apt install dovecot-core dovecot-imapd dovecot-pop3d
```

Check Dovecot version:

```
dovecot --version
```

- Enabling IMAP/POP3 Protocol

```
sudo vi /etc/dovecot/dovecot.conf
protocols = imap pop3
```

```
sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config
```

Q : SASL Quite mode
LLL : Print responses in LDIF format without comments
-Y : SASL mechanism
-H : LDAP URL
-b : basedn

```
dn: cn=config
dn: cn=module{0},cn=config
dn: cn=schema,cn=config
dn: cn={0}core,cn=schema,cn=config
dn: cn={1}cosine,cn=schema,cn=config
dn: cn={2}nis,cn=schema,cn=config
dn: cn={3}inetorgperson,cn=schema,cn=config
dn: olcDatabase={-1}frontend,cn=config
dn: olcDatabase={0}config,cn=config
dn: olcDatabase={1}mdb,cn=config
```

View Information

This is what the dc=vusna,dc=com DIT looks like:

```
$ ldapsearch -x -LLL -H ldap:/// -b
dc=vusna,dc=com dn

dn: dc=vusna,dc=com

dn: cn=admin,dc=vusna,dc=com
```

verify the authentication with ldapwhoami

```
Authenticating as principal root/admin@vusna.com with password.

kadmin.local: addprinc sna/admin

WARNING: no policy specified for sna/admin@vusna.com; defaulting to no policy

Enter password for principal " sna/admin@vusna.com ":

Re-enter password for principal " sna/admin@vusna.com ":

Principal "sna/admin@vusna.com" created.

kadmin.local: quit
```

Create Admi Principal

6 - Set admin p cipal with appropriate Access Control List (ACL) permissions. Th
are nfigur n the /etc/krb5kdc/kadm5.acl file:

```
sna/admin@vusna.com      *
```

The a e will grant all privileges to any admin instance of a principal. See the kad
deta

- R art the krb5-admin-server for the new ACL to take affect:

```
sudo systemctl restart krb5-admin-server.service
```

8 - The new user principal can be tested using the kinit utility:

```
$ kinit sna/admin
Password for sna/admin@vusna.com:
```

KDC Installation

1 - Install the Kerberos Packages

- krb5-kdc
- krb5-admin-server
- sudo apt install krb5-kdc krb5-admin-server

2 - Create the new realm with the kdb5_newrealm utility

- sudo krb5_newrealm

3 - you can reconfigure Kerberos from scratch, if need to change the realm name. Following commands can be used for this purpose.

- sudo dpkg-reconfigure krb5-kdc

KDC Configuration

4 - Create the first principal.

```
sudo kadmin.local
Authenticating as principal root/admin@vusna.com with password.
kadmin.local: addprinc sna
WARNING: no policy specified for sna@vusna.com; defaulting to no policy
Enter password for principal "sna@vusna.com":
Re-enter password for principal " sna@vusna.com ":
Principal "sna@ vusna.com" created.
kadmin.local: quit
```

Create Admin Principal

5 - Create an admin principal and assign permissions.

```
$ sudo kadmin.local
Authenticating as principal root/admin@vusna.com with password.
kadmin.local: addprinc sna/admin
WARNING: no policy specified for sna/admin@vusna.com; defaulting to no policy
Enter password for principal " sna/admin@vusna.com ":
Re-enter password for principal " sna/admin@vusna.com ":
Principal "sna/admin@vusna.com" created.
kadmin.local: quit
```

Create Admin Principal

6 - Set admin principal with appropriate Access Control List (ACL) permissions. The permissions are configured in the /etc/krb5kdc/kadm5.acl file:

```
sna/admin@vusna.com      *
```

The above will grant all privileges to any admin instance of a principal. See the kadm5.acl manpage for details.

7 - Restart the krb5-admin-server for the new ACL to take affect:

```
sudo systemctl restart krb5-admin-server.service
```

8 - The new user principal can be tested using the kinit utility:

```
$ kinit sna/admin
Password for sna/admin@vusna.com:
```

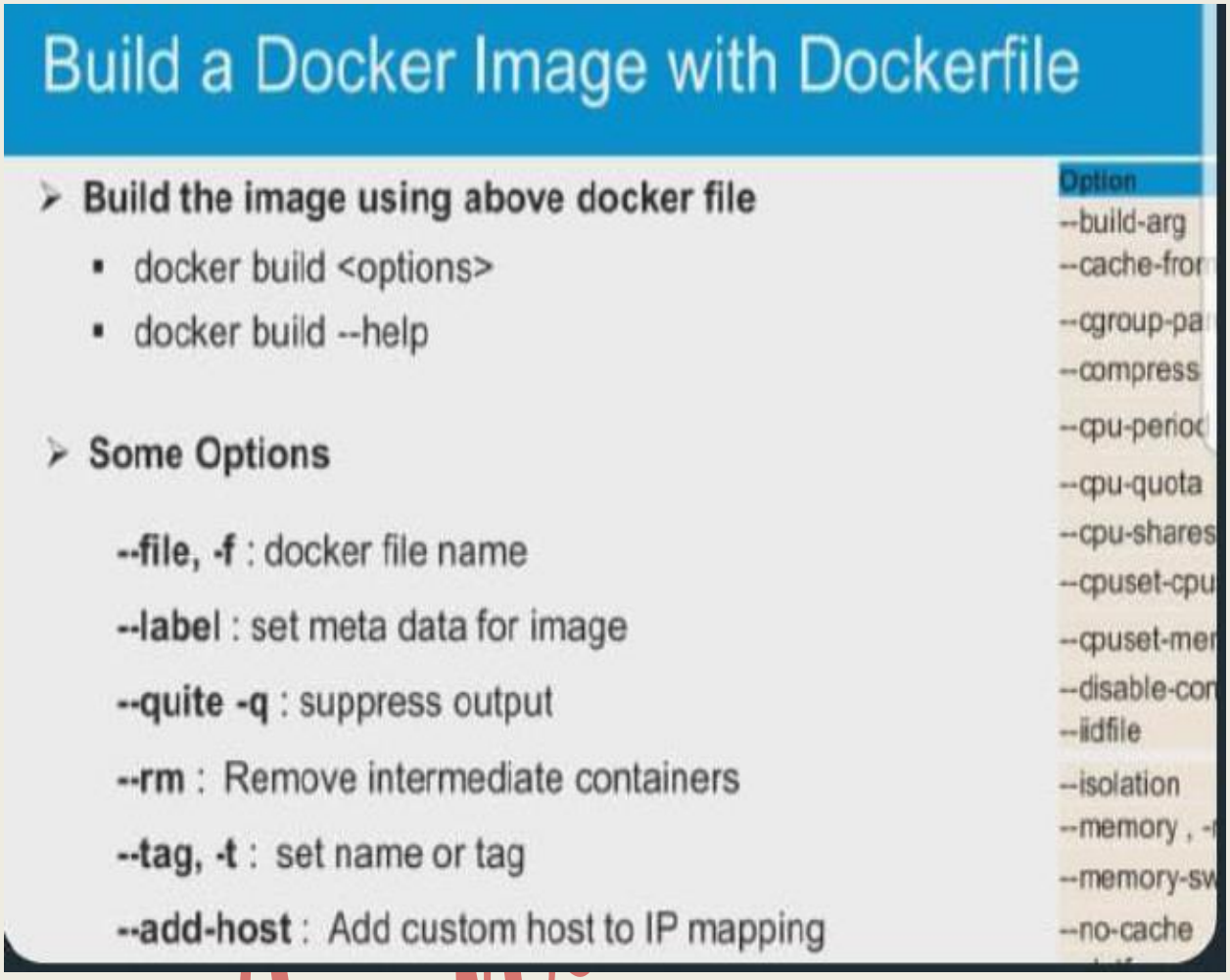
named-checkconf /etc/bind/named.conf.options

1) mkdir /etc/bind/zones
2) cp /etc/bind/db.local /etc/bind/zones/vusna.com
3) vi /etc/bind/zones/vusna.com

VU[illegible].COM Zone Configuration

```
$TTL    604800
@       IN      SOA     vusna.com. root.vusna.com. (3; 604800; 86400; 2419200
@       IN      NS      bindserver.vusna.com.
bindserver      IN      A       192.168.2.2
web     IN      A       192.168.2.3
dhcp    IN      A       192.168.2.4
db      IN      A       192.168.2.5
elk     IN      A       192.168.2.6
```

Maha R



Maha

----------**THE END**----------

May Allah grant you success, ease your efforts, and bless you with wisdom and perseverance. Keep your faith strong, for with prayer, every challenge becomes an opportunity. I hope this file helps you a lot, and please remember me in your prayers.

Best wishes for Finals!
Maha ☺

